

Lecture 9

Sampling, Experiments and Causation

Review

1 What is bias?

When a sampling method's estimates miss the parameter in the same direction every time - an accuracy problem.

2 What is standard error?

How much an estimate varies from one sample to the next - a precision problem.

3 What is random sampling?

Using chance, not a person, to decide which units get into the sample.

4 What is a simple random sample (SRS)?

Every possible group of n units has the same chance of being the sample.

Review

5 What is a convenience sample?

A sample of whoever is easiest to reach.

6 What is sampling with replacement?

A picked unit goes back into the pool, so it can be picked again.

7 What is sampling without replacement?

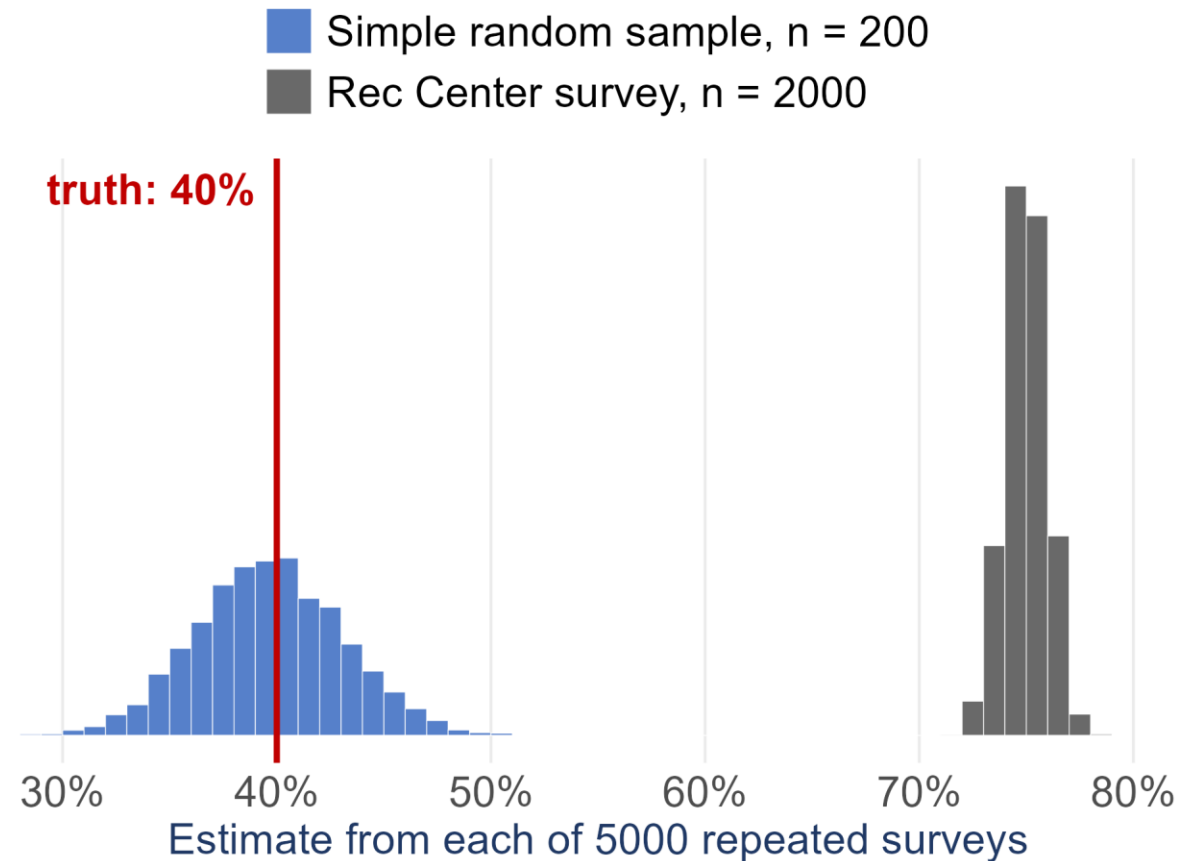
A picked unit is set aside, so it can be chosen only once.

8 What is a voluntary response sample?

A sample of people who choose themselves by responding - usually the ones with strong opinions.

A bigger sample does not fix bias

- Suppose 40% of UI students exercise at least three times a week
- Blue: 5000 simple random samples of 200 students
- Grey: 5000 Rec Center surveys of 2000 students
- The Rec Center survey is more **precise** - a smaller standard error
- and every one of its estimates misses the truth: **bias**
- More data buys precision. Only a **better design** buys accuracy



In the news: how a national poll samples

- State of the Facts 2026 (AP-NORC Center with USAFacts) asked U.S. adults how much they trust government information and AI
- 1,036 adults, July 29 – August 10, 2026: 1,017 online and 19 by phone, in English or Spanish
- Drawn from AmeriSpeak, a **probability-based panel**: NORC randomly selects U.S. addresses, then recruits those households by mail, phone and in person
- You cannot sign yourself up - **chance decides** who is invited. Compare the opt-in panels from last week
- Margin of sampling error: ± 4.1 percentage points
- But only **10.3%** of invited panel members completed this survey (AAPOR response rate 3)
- One finding: only 37% say it is easy to know whether the information they see is true

Read more: [Project page](#) · [Full report \(PDF; methods on p. 42\)](#) · [How AmeriSpeak recruits](#)

Sources: AP-NORC Center for Public Affairs Research and USAFacts (2026); NORC, About AmeriSpeak

In the news: light at night, over time

- UK Biobank: about 500,000 UK adults aged 40–69 who agreed to join in 2006–2010, then followed through their health records
- 2013–2015: 103,615 of them wore a wrist sensor with a light meter for 7 days
- Removed: unreliable sensor data, 2,770 night-shift workers, anyone with heart disease already, anyone without a later heart MRI → 11,071 people
- About 3 years later each had an MRI of their heart; hospital and death records were followed for up to 10 years
- **Longitudinal**: the same people measured more than once - night light first, the heart later
- **Not random**: the people who agreed to join are mostly White and healthier than the UK population (the authors say so)
- Result: more light above 3 lux at night went with thicker heart walls and weaker pumping. Did the light cause it? We come back to that later today

Read more: [ESC press release](#) · [Dai et al. \(2026\), European Heart Journal](#)

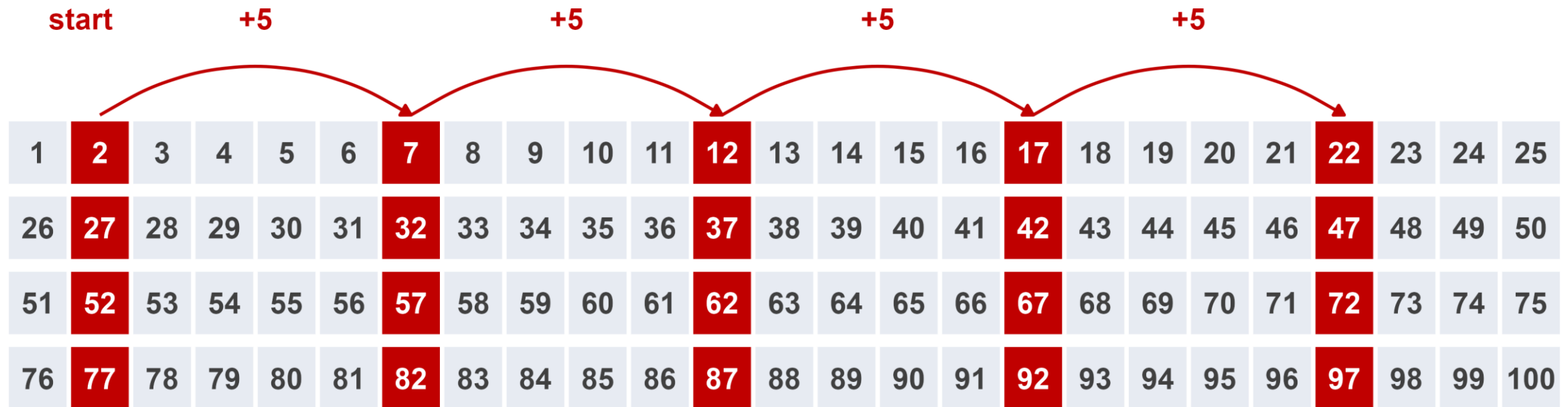
Sources: Dai, Dai, Heianza and Qi (2026), European Heart Journal; ESC press release (2026)

Systematic sample

systematic sample

pick a random starting point among the first k units of an ordered list, then take every k th unit after it

$k = N/n = 100/20 = 5$. Random start = 2, so we take list positions 2, 7, 12, 17, ... , 97



Works like an SRS - unless the list has a repeating pattern that lines up with k

Stratified random sample

strata

groups of units that are alike, formed BEFORE sampling - class year, region, age group

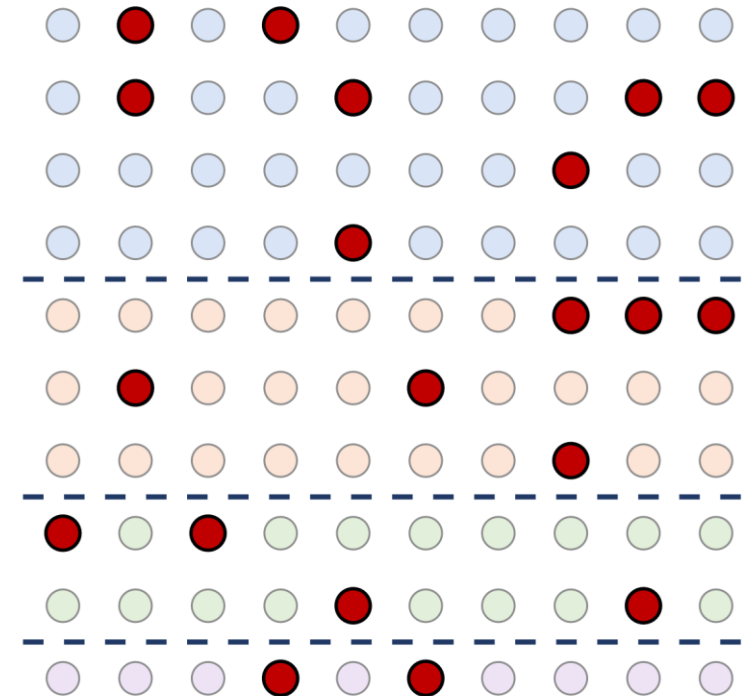
stratified random sample

a separate SRS inside every stratum, combined into one sample

proportional allocation

each stratum gets its share of n : 40, 30, 20, 10 students $\rightarrow$ 8, 6, 4, 2

Every group is guaranteed a place - no more 2 juniors by bad luck



● First-year ● Junior
● Sophomore ● Senior

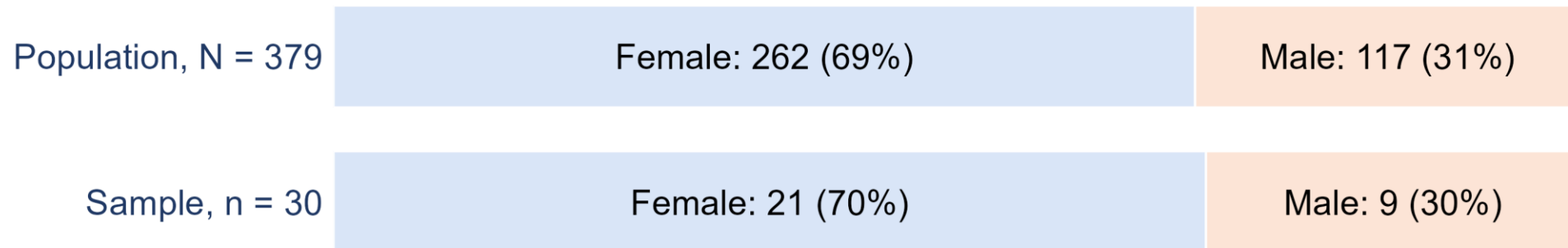
Proportional allocation with real data

heights.csv (on the course Data page) has $N = 379$ people: 262 female and 117 male. Stratify by sex for a sample of $n = 30$:

female: $30 \times 262/379 = 20.74$, so 21

male: $30 \times 117/379 = 9.26$, so 9

Then an SRS of 21 of the women, and a separate SRS of 9 of the men



Cluster sample

cluster

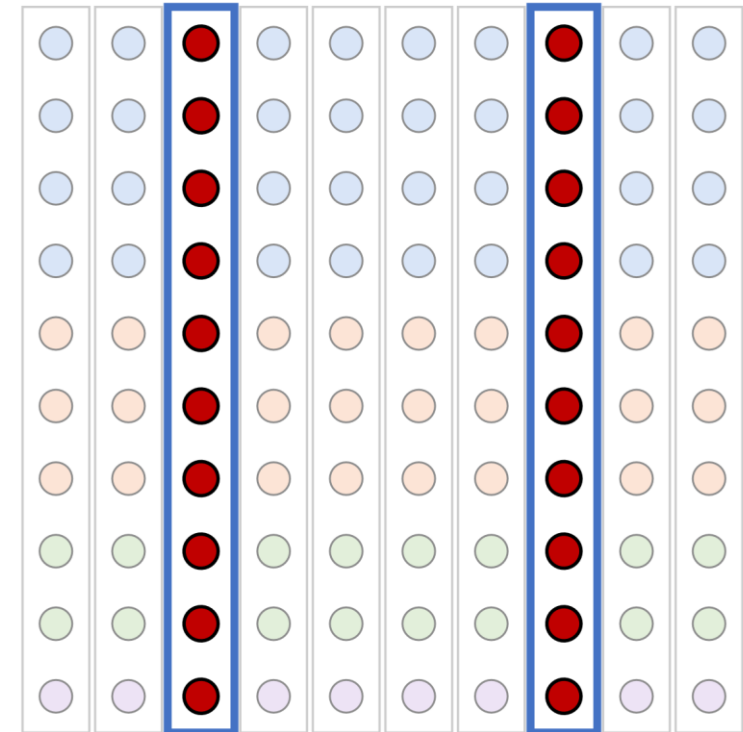
a group of units that occurs naturally - a dorm, a classroom, a city block

cluster sample

an SRS of clusters, then measure EVERY unit in the clusters chosen

Only needs a list of clusters, not of every person - and far less travel

Works best when each cluster is a mini-population: mixed inside, like each column here



● First-year ● Junior
● Sophomore ● Senior

Two-stage cluster sample

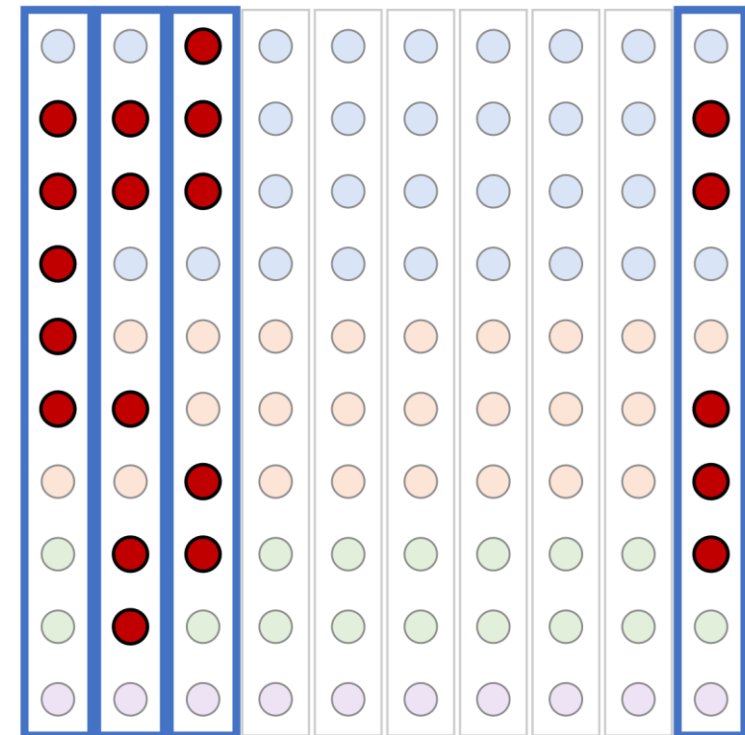
two-stage cluster sample

stage 1: an SRS of clusters. Stage 2: an SRS of units inside each chosen cluster

Here: 4 of the 10 columns, then 5 students in each, $n = 20$

Exit polls work this way: sample polling places, then interview every n th voter leaving each one

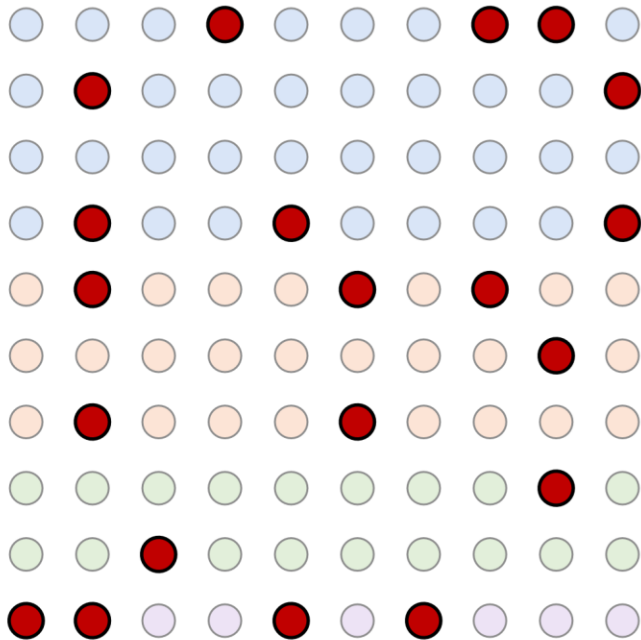
National surveys such as NHANES stack it all: strata, clusters inside strata, households inside clusters



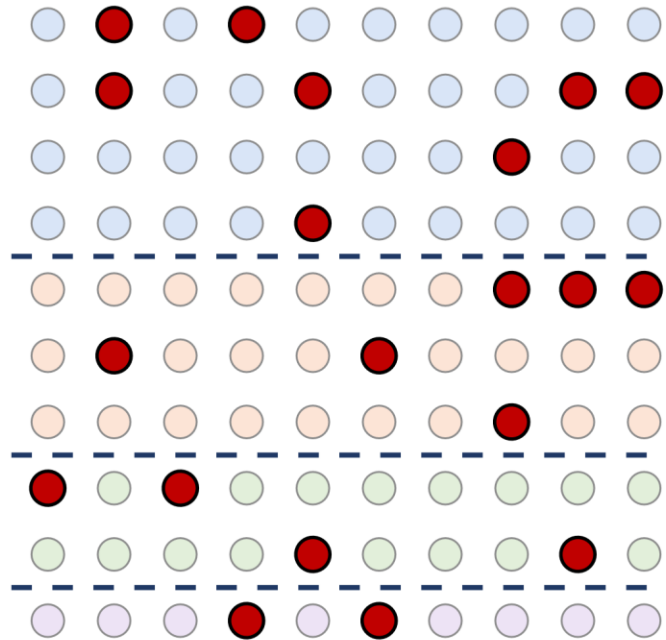
● First-year ● Junior
● Sophomore ● Senior

Strata vs. clusters

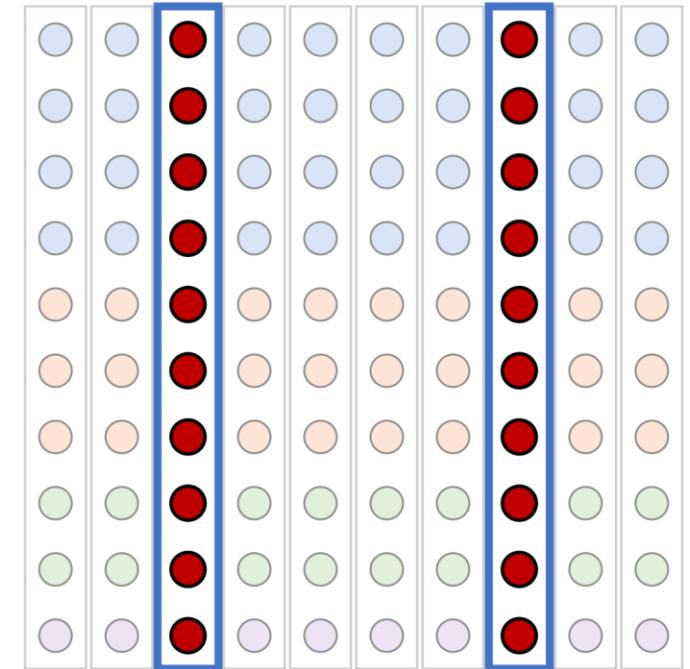
Simple random



Stratified



Cluster



● First-year ● Sophomore ● Junior ● Senior

Strata: alike inside - sample from EVERY one. Clusters: mixed inside - sample SOME, whole.

Two kinds of error

sampling error

the natural gap between a sample and the population, caused by which units happened to be chosen

every sample has some - a larger sample makes it smaller

nonsampling error

errors that do not come from chance - biased methods, bad questions, data-entry mistakes, faulty instruments

a larger sample does not make these smaller

sampling bias

some members of the population are less likely to be chosen than others

convenience and voluntary response samples, or a frame that leaves people out

Three common nonsampling errors

Even a perfectly random design can end up biased:

undercoverage

the sampling frame leaves part of the population out
students who are not on the list you sampled from

nonresponse

people who are chosen cannot be reached, or refuse to answer
a problem when the people who do not answer differ from those who do

response bias

the answers themselves are wrong
leading or confusing questions, sensitive topics, wanting to please the interviewer